

HTML 5 Foundations



Learning Objectives

- Understand the relationship between the Browser, the Server, and HTML.
- Set up a professional development environment with VS Code.
- Master Semantic HTML to ensure your sites are accessible and SEO-friendly.
- Learn how to nest elements correctly (Parent-Child relationships).

Week One

Module One



What is HTML?



What is HTML?

- HTML stands for **Hyper Text Markup Language**, which is the most widely used language on the web to develop web pages
- The Analogy: If a website were a house, HTML is the framing and the walls. It's not the paint (CSS) or the plumbing (JS); it's the structure.
- HTML is not a programming language; it is a markup language. It "marks up" plain text so a browser knows what it is.



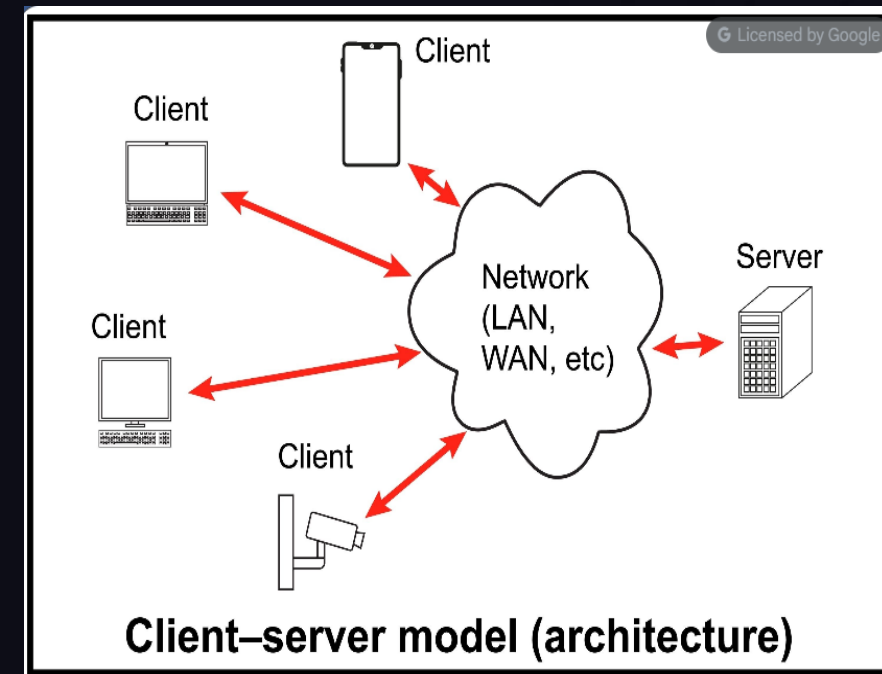
Why is HTML So Important?

- **The Foundation:** Every website on the planet uses HTML.
- **Accessibility:** Proper HTML helps screen readers describe the page to visually impaired users.
- **SEO (Search Engines):** Google "reads" your HTML to decide what your site is about.
- It works on all devices the same way.



How the Web Works?

- **The Client (You):** Your browser asks for a file.
- **The Server:** A computer far away sends the HTML file back.
- **The Browser's Job:** It reads the tags and renders (draws) the page for you.



Basic HTML Structure

```
<html>

  <head>

    <title>Page title</title>

  </head>

  <body>

    <h1>This is a heading</h1>

    <p>This is a paragraph.</p>

    <p>This is another paragraph.</p>

  </body>

</html>
```



Headings & Paragraphs



Headings and Paragraph

- **Headings (<h1> to <h6>):**

- <h1> is your page title. (Search engines love this!)
- <h2> - <h6> are for sub-sections.

- **Paragraphs (<p>):**

- Used for all your regular text.
- Never choose a heading tag based on the text size; choose it based on importance.



HTML Lists

- **Unordered ():** Bullet points (shopping lists, navigation).
- **Ordered ():** Numbered lists (recipes, rankings).
- **List Items ():** The actual items inside the lists..



HTML Links

The <a> Tag: This is why it's called a "web."

- **Attributes:** Tags can have extra info.
- href: The destination URL.

Example: `<a href="https://google.com">Click me!</a>`



Adding Visuals

(Images & Videos)



HTML Images

- The **** Tag: It's a "self-closing" tag.

Attributes:

- **src:** The path to the image file.
- **alt:** The description (Crucial for accessibility!).

Why Alt Text? If the image fails to load, the user still knows what was supposed to be there.



HTML Video & Media

The **<video>** Tag: Allows you to host video files directly on your site.

Essential Attributes:

- **src:** The path to the video file.
- **controls:** Crucial. Adds the play/pause/volume buttons (otherwise it's just a still image!).
- **width/height:** Sets the player size.

The **YouTube Method:** Mention `<iframe>` (The "Embed" button on YouTube) as the industry-standard way to add video without slowing down your own server.



HTML Tables & Forms



HTML Video & Media

- Tables are for data sets (schedules, prices, scores).
- Never use tables for your website layout (that's what CSS is for). Use them only for tabular information.

The Table Family:

- `<table>`: The main wrapper.
- `<tr>`: Table Row.
- `<th>`: Table Heading (automatically bolds the text).
- `<td>`: Table Data (the standard cells).



HTML Forms

The **<form>** Tag: The container for user input.

The **"Action"** & **"Method"**:

- action: Where the data goes (the "mailbox").
- method: How it gets there (usually "POST" for security).

Why Forms Matter: This is how users sign up, log in, search, and buy products.



The Input Family

The **<input>** Tag: A versatile, self-closing tag. Its behavior changes based on the type attribute:

- **type="text"**: A standard single-line box.
- **type="email"**: Checks for an "@" symbol automatically.
- **type="password"**: Hides the characters as you type.
- **type="submit"**: The button that sends the form.
- **<textarea>**: For longer, multi-line messages (like a "Bio" section).



Labels & Accessibility

- The **<label>** Tag: Every input must have a label.
- The **for** Attribute: Links the label to the input's id.

Why? It makes the form accessible to screen readers and allows users to click the text to select the input box.



Semantic HTML

The Old Way: Using `<div>` for everything.

The Modern Way:

- `<header>` (Top nav/logo)
- `<main>` (Primary content)
- `<section>` (Logical groupings)
- `<footer>` (Bottom info)

Benefit: Better for SEO and much easier for other developers to read your code.



Summary

- HTML is Structure. It defines content and organization rather than visual style.
- Proper use of Headings creates a logical map for SEO and screen readers.
- Tags like `<a>` and `<img>` require attributes (`href`, `src`, `alt`) to function and ensure accessibility.
- Tables organize complex information, while Forms (inputs, labels, and buttons) allow for user interactivity.
- Using meaningful tags like `<main>`, `<nav>`, and `<article>` ensures clean, professional, and accessible code.



Home Exercise 1: The Digital Resume

Goal: Build a resume using only the tags we learned.

Requirements:

- Header & Biodata
- Profile Statement
- Educational Background
- Work Experience
- Technical Skills
- "Work With Me" (The Form)
- The Footer



Check the drive folder for detailed instructions.

